

$$B_t = \underline{2(W_3) + 3}$$

B_t is normal ✓

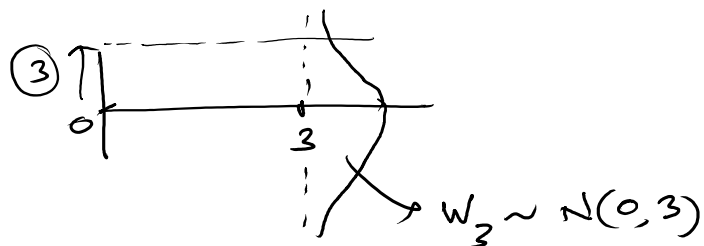
$$\mathbb{E}[B_t] = \mathbb{E}[2W_3 + 3] =$$

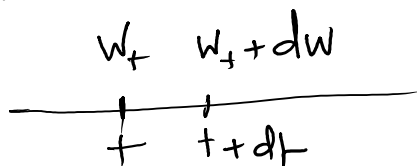
$$= \mathbb{E}[2W_3] + \mathbb{E}[3]$$

$$= 2\mathbb{E}[W_3] + 3 = \underline{\underline{3}}$$

$$\mathbb{V}[B_t] = \mathbb{V}[2W_3 + 3] = \mathbb{V}[2W_3] = 4 \times \mathbb{V}(W_3) = 4 \times 3 = 12$$

$$B_t \sim N(3, 12)$$

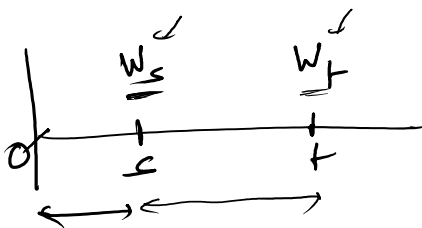
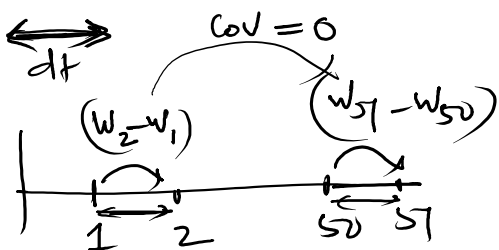




$$dw \sim N(0, dt)$$

$$\text{std. dev} = \sqrt{dt}$$

①



$$w_s \sim N(0, s)$$

$$w_t \sim N(0, t)$$

$$w_t - w_s \sim N(0, t-s)$$

$$\begin{aligned} \text{Cov}(w_s, w_t) &, s \leq t \\ &= \text{Cov}(w_s, w_s + (w_t - w_s)) \\ &= \text{Cov}(w_s, w_s) + \text{Cov}(w_s, w_t - w_s) \\ &= s + 0 = s \quad (\text{overlapping region}) \end{aligned}$$

$$\text{Cov}(w_s, w_t) = \min(s, t)$$

$$\text{Corr}(w_s, w_t) = \begin{cases} \frac{s}{\sqrt{st}} & , s \leq t \\ \frac{t}{\sqrt{st}} & , s \geq t \end{cases}$$

$$= \begin{cases} \sqrt{\frac{s}{t}} & , s \leq t \\ \sqrt{\frac{t}{s}} & , s \geq t \end{cases}$$

$$\text{Corr} = \sqrt{\frac{\min(s, t)}{\max(s, t)}}$$

$$P(\underline{W_1 + W_2} > 2)$$

$$\begin{matrix} x_1 \\ x_2 \end{matrix} \sim N(0, 1), \rho = 0$$

$$P(x_1 > x_2) \quad \text{HW}$$

$$B = W_1 + W_2$$

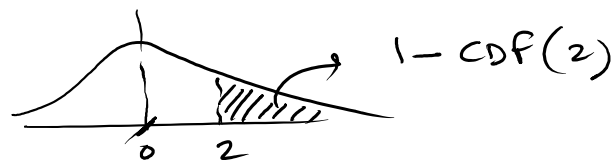
$$E[B] = 0$$

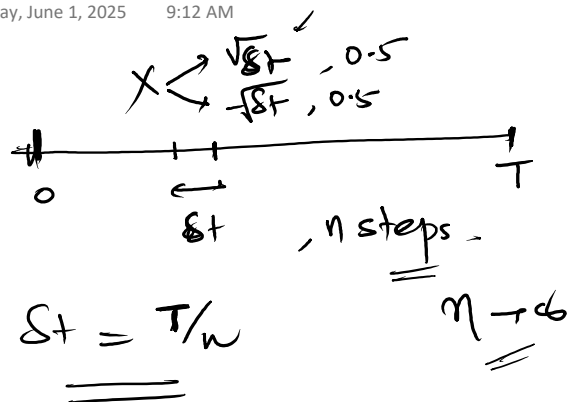
$$V(B) = V(W_1) + V(W_2) + 2 \cdot \text{Cov.}(\underline{W_1}, \underline{W_2})$$

$$= 1 + 2 + 2 \times 1 = 5$$

$$\text{Cov}(W_s, W_t) = \min(s, t)$$

$$B \sim \underline{N(0, 5)}, \quad P(\underline{B} > 2)$$





$$\begin{aligned}
 \mathbb{E}(x) &= 0 \\
 \mathbb{V}(x) &= (\sqrt{\delta t} - 0)^2 \cdot (0.5) \\
 &\quad + (-\sqrt{\delta t} - 0)^2 \cdot 0.5 \\
 &= \delta t
 \end{aligned}$$

$$Y_T = Y_0 + \underbrace{X_1 + X_2 + \dots + X_n}$$

$$\mathbb{E}[Y_T] = Y_0 = 0$$

$$\mathbb{V}[Y_T] = n \times \delta t = T$$

$$Y_T \underset{\eta \rightarrow \infty}{\overset{\text{CLT}}{\sim}} \mathcal{N}(0, T) \Rightarrow W_T$$

$$w_1 = 0.5, w_2 = 1, w_3 = -2, w_5 = ??$$

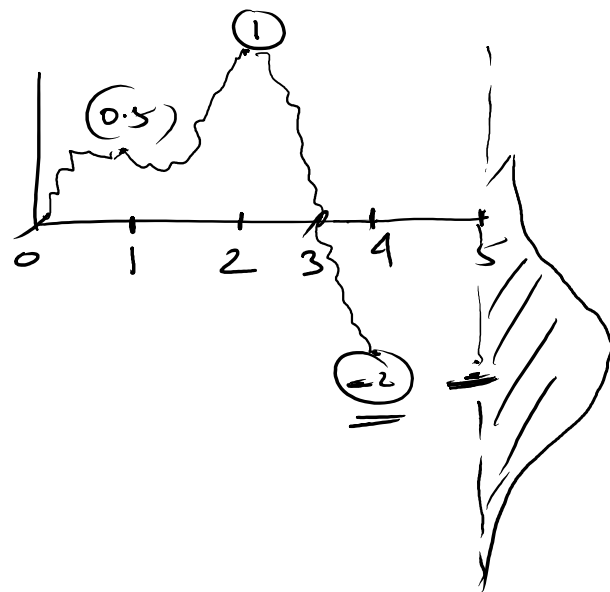
$$w_5 \sim N(0, 5)$$

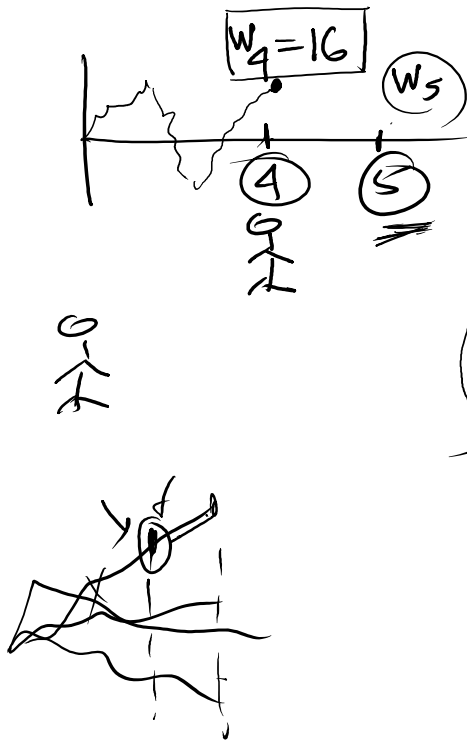
$$(w_5 | w_3 = -2)$$

$$(w_5) - w_3 \sim N(0, 2)$$

$$w_5 | w_3 \sim (w_3) + N(0, 2) \leftarrow$$

$$w_5 | w_3 \sim N(-2, 2)$$





$$W_5 | W_4 = 16$$

$$W_5 = 16 + (W_5 - W_4)$$

$$W_5 - W_4 \sim N(0, 1)$$

$$W_5 | W_4 \sim N(16, 1)$$

$$W_5 \sim N(0, 5)$$

W_t is F_t
adapted

$$\mathbb{E}[W_t | W_s]$$

$$= \mathbb{E}[W_s + (W_t - W_s) | W_s]$$

$$= \mathbb{E}[W_s | W_s] + \mathbb{E}[\underbrace{(W_t - W_s)}_{\text{independent}} | W_s]$$

$$= W_s + 0 = W_s$$

$$W_t - W_s \sim N(0, t-s)$$

$$\mathbb{E}[W_t | W_s] = W_s \rightarrow W_t \text{ (martingale)}$$

$$\underline{W_1(t)} = \frac{1}{\sqrt{c}} (\underline{W(ct)}) , \quad c \rightarrow \text{scalar}$$

✓ W_1 is normal ✓

$$✓ \mathbb{E}[W_1] = \frac{1}{\sqrt{c}} \times 0 = 0$$

$$✓ \mathbb{V}[W_1] = \frac{1}{c} \times ct = t$$

$$W_1(t) \sim N(0, t) = W_t$$

(Scaling)

$$W_2(t) = t W_{(1/t)}$$

✓ $W_2(t)$ is normal

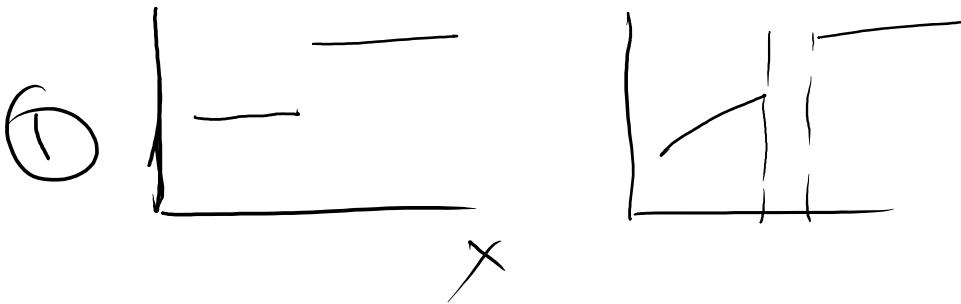
$$\mathbb{E}[W_2(t)] = t \times 0 = 0$$

$$\mathbb{V}[W_2(t)] = t^2 \times \frac{1}{t} = t$$

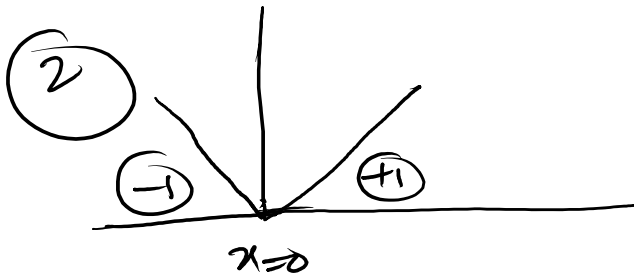
$$W_2(t) \sim N(0, t)$$

$$= W_t$$

(Inversion)

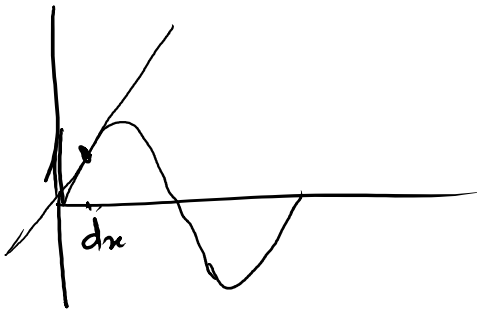


discontinuous
→ non-differentiable



$y = |x|$
not differentiable
at $x=0$

③ squiggly/jagged.



($\sin x$)

$\frac{dx}{dx}$

$\frac{dL}{dt}$

$$dw \sim N(0, dt)$$

$$dw = \underline{\sqrt{dt}} \cdot Z, \quad Z \sim N(0, 1)$$

$$\frac{dw}{dt} = \left(\frac{Z}{\sqrt{dt}} \right)$$

$dt \rightarrow 0$ slope explodes

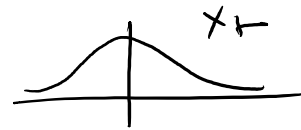
if $\underline{x}_t \sim \underline{N}(\underline{\mu}, \underline{\sigma}^2)$

$\underline{y}_t = e^{\underline{x}_t} \sim \underline{\text{lognormal}}(\underline{\mu}, \underline{\sigma}^2)$

$\rightarrow \underline{E}[\underline{y}_t] = \underline{e^{(\mu + \frac{1}{2}\sigma^2)}}$

$\ln y_t = x_t \sim N(\mu, \sigma^2)$

$\underline{V}[\underline{y}_t] = ??$



$$\textcircled{1} S_t = \underline{\mu t} + \sigma W_t$$

$$S_{t_2} - S_{t_1} = (\mu t_2 + \sigma W_{t_2}) - (\mu t_1 + \sigma W_{t_1})$$

$$= \underline{\mu(t_2 - t_1)} + \sigma \underline{(W_{t_2} - W_{t_1})}$$

$$= \mu(t_2 - t_1) + \sigma \cdot N(0, t_2 - t_1)$$

independent

W_{t_2}

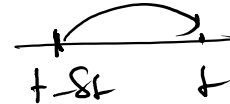
$\mu(t) \cdot \underline{dt}$

$$\textcircled{2} S_t = e^{\mu t + \sigma W_t}$$

$$S_{t_2} - S_{t_1} = e^{\mu t_2 + \sigma W_{t_2}} - e^{\mu t_1 + \sigma W_{t_1}}$$

$$(W_{t_2} - W_{t_1})$$

not independent



$$\log(S_t) = \mu t + \sigma W_t$$

$$\log(S_{t_2}) - \log(S_{t_1}) = \mu(t_2 - t_1) + \sigma(W_{t_2} - W_{t_1})$$

$$\log(S_t) - \log(S_{t-\delta t}) = \underline{\underline{\mu \delta t}} + \underline{\underline{\sigma Z \sqrt{\delta t}}}$$

$$\underline{\underline{\log(\text{return})}} \sim N(\mu \delta t, \sigma^2 \delta t)$$

$$\begin{aligned} \text{mean} &= \mu \delta t \\ \text{std. dev} &= \sigma \sqrt{\delta t} \end{aligned}$$

$$A_t - A_s \sim U(- (t-s), 5(t-s))$$

$$\mathbb{E}[A_t - A_s] = \frac{-(t-s) + 5(t-s)}{2}$$

$$= 2(t-s)$$

$$\mathbb{E}[A_t - A_0] = 2t$$

$(A_t - 2t) \rightarrow \text{martingale}$

$$\mathbb{E}[A_t - 2t] = \mathbb{E}[A_t] - 2t$$

$$= A_0 + 2t - 2t = A_0$$

$$X_t \sim U(a, b)$$

$$\mathbb{E}[X_t] = \left(\frac{a+b}{2}\right)$$



$$X_t = e^{\underline{\underline{-\frac{1}{2}\sigma^2 t + \sigma W_t}}}$$

$$Y_t = \underline{\underline{-\frac{1}{2}\sigma^2 t + \sigma W_t}}$$

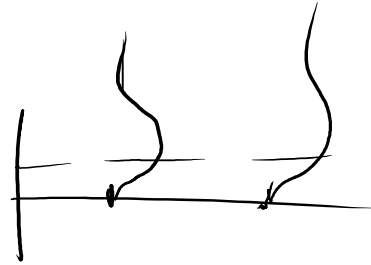
$$\underline{Y_t} \sim N\left(\underline{\underline{-\frac{1}{2}\sigma^2 t}}, \sigma^2 t\right)$$

$$X_t = e^{Y_t} \sim \text{lognormal}\left(\underline{\underline{-\frac{1}{2}\sigma^2 t}}, \sigma^2 t\right)$$

$$\mathbb{E}[X_t] = e^{-\frac{1}{2}\sigma^2 t + \frac{1}{2}\sigma^2 t} = e^0 = 1$$

exponential martingale

$$\underline{\underline{X_t}} = \underline{\underline{X_0}} \cdot \underline{\underline{e^{-\frac{1}{2}\sigma^2 t + \sigma W_t}}} \Rightarrow \mathbb{E}(X_t) = \underline{\underline{X_0}}$$



$$B_t = \underline{W_t^2 - t}$$

$$\begin{aligned} \mathbb{E}[B_t] &= \mathbb{E}(W_t^2) - t \\ &= \mathbb{E}[t \cdot Z^2] - t \\ &= t \mathbb{E}(Z^2) - t = t - t = 0 \end{aligned}$$

$$\begin{aligned} Z &\sim N(0,1) \\ \underline{Z^2} &\sim \text{chisq}(1, 2) \\ \underline{Z_1^2 + Z_2^2 + \dots + Z_k^2} &\sim \text{chisq}(k, 2k) \end{aligned}$$

$$W_t = (\sqrt{t}) \cdot Z$$

$$\begin{aligned} X_t &= (W_t^2 - t) - (W_s^2 - s) \\ &= [W_s + (W_t - W_s)]^2 - t - (W_s^2 - s) \\ &= \cancel{W_s^2} + \underline{(W_t - W_s)^2} + 2 \cdot W_s \cdot (W_t - W_s) - t - \cancel{(W_s^2 - s)} \\ &= \underline{(t-s) \cdot Z^2} + 2 \cdot \underline{W_s} \cdot \underline{(W_t - W_s)} - (t-s) \end{aligned}$$

$$\mathbb{E}[X] = (t-s) - (t-s) = 0$$

if X & Y are independent

$$\begin{aligned} \mathbb{E}(XY) &= \mathbb{E}(X) \cdot \mathbb{E}(Y) \\ &= 0 \times 0 = 0 \end{aligned}$$

$$\begin{aligned} \mathbb{E}(Z^2) &= 1, \quad \underline{V(Z)} = \underline{\mathbb{E}(Z^2)} - (\underline{\mathbb{E}(Z)})^2 \\ &\Rightarrow \underline{(1)} = \underline{(\text{??})} - 0 \end{aligned}$$